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Geoeconomic Risk Exposure and Investments: EU firms' Response to Russia's Invasion of Ukraine

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Abstract

This paper studies the geoeconomic risk exposure of firms in the European Union (EU) and estimates its impact on investments. The first novelty of this paper is the construction of the *GRE*: a new text-based index using corporate earnings calls, allowing for the identification of geoeconomic risk exposure at the firm-level. The *GRE* intuitively tracks major geoeconomic events, displays expected geographic and sectoral heterogeneity, and shows that in periods of elevated uncertainty, a mix of geoeconomic issues become more salient. The paper's second contribution is to estimate the impact of Russia's invasion of Ukraine on highly exposed EU firms' investments. Firms with above-median *GRE* increase investment by 12% following the start of the war—roughly double firm's average annual growth rate in investments. Utilities firms exhibit a stronger response to geoeconomic exposure, increasing investment by roughly 55% more than firms in other sectors, whereas exposed financial firms reduce investment by about 45% relative to non-financial firms. In light of these results, I discuss how geoeconomic risk may operate through 2 channels: one that prompts firms to increase investments to diversify away from risky countries, and another that heightens the uncertainty firms are facing, and thereby reduces investments.

Keywords: geoeconomics; firm investment; text-as-data; Russia–Ukraine war.
JEL codes: D22; E22; F51; F62; C55.

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1 Introduction

This paper contributes to better understanding how geoeconomics impacts firms in the European Union (EU) by constructing a new firm-level index of geoeconomic risk exposure (GRE) and estimating how it shapes firms' investments following Russia's invasion of Ukraine.

Trebesch and Mohr (2025) define geoeconomics as the study of how political rivalry, including direct conflict, shapes economic policies and outcomes, and vice versa. Firms sit at the centre of this renewed great power rivalry and the associated increased use of economic warfare. Due to the internationalisation of supply chains and substantial cross-border operations, firms have been the targets of economic coercion through the "weaponisation" of their dependencies (Drezner et al., 2021). A recent example is the U.S. Treasury's sanctions on Russia's oil majors Rosneft and Lukoil in October 2025 to ramp up pressure on the Russian government ahead of peace negotiations with Ukraine. At the same time, firms headquartered in the sanction-imposing country may also be adversely affected as a side effect of these actions or due to potential retaliation. The debate is particularly salient in the EU, where firms' long-standing reliance on cheap Russian energy imports was revealed as a strategic vulnerability after Russia's invasion of Ukraine. This dependency shaped the debate over whether to impose a gas embargo, given the potential consequences of a sudden cut-off in Russian gas for EU firms and broader macroeconomic outcomes (see Moll et al. (2023) for a review of the debate in Germany). A key feature of this debate was the adaptation mechanism (substitution away from Russian gas), which ultimately limited the downturn in Germany to a mild recession. This paper provides further evidence on firms' responsiveness to geoeconomic shocks by examining the investment response of more exposed EU firms.

I first address the challenge of measuring geoeconomic risk exposure at the firm level—for which exposure to sanctions, export controls, geopolitical events, and related threats is hard to observe and track over time—by using transcripts from corporate earnings calls. In each transcript, I compute the share of sentences that mention geoeconomic risks by applying a keyword-based search query covering geoeconomic instruments and events, following established text-as-data approaches (Hassan et al. (2019); Caldara and Iacoviello (2022); Niepmann and Shen (2024); Clayton et al. (2025a)). This exercise yields a firm-quarter

index of geoeconomic risk exposure, the *GRE*. Four stylised facts emerge from the index construction. First, the *GRE* demonstrates that firm discussions, on aggregate, are responsive to major geoeconomic events, with spikes occurring during the Iraq war, Russia's annexation of Crimea, the 2018 US-China trade war, Russia's full-scale invasion of Ukraine, and Liberation Day. Second, periods of elevated geoeconomic risk coincide with a broadening in the mix of instruments discussed, consistent with firms facing multiple channels of geoeconomic risk at once. Third, sectoral exposure is highly heterogeneous, with large differences across industries, likely related to their international exposure. Fourth, geoeconomic risks are mentioned at least once in roughly a quarter of all earnings calls, but repeated mentions within a single call are rare.

The paper then exploits Russia's invasion of Ukraine to estimate the effect of geoeconomic risk exposure on firm investments using a differences-in-differences design based on the *GRE*. Firms are assigned to high-exposure (treatment) or low-exposure (control) groups depending on whether their *GRE* in the first year of the Russo-Ukrainian war lies above or below the median. The *GRE* for the two groups evolves in parallel prior to the invasion, but rises sharply for the high-exposure group thereafter. Investment likewise follows parallel trends before the invasion; afterwards, investment increases for high-exposure firms and declines for low-exposure firms.

The main results empirically confirm that firms with high geoeconomic risk exposure increase investment by 12% following Russia's invasion of Ukraine—about twice the average annual investment growth rate of those firms. The effects are heterogeneous across sectors: highly exposed utilities firms raise investment by over 55% relative to other firms, whereas highly exposed financial firms reduce investment by about 45% relative to non-financial firms. High geoeconomic risk exposure does not affect other firm-level variables except for slightly increasing its cost of goods sold, and does not hold for non-EU firms. The positive effect on investments for high exposure firms holds up to a series of robustness tests. I formally confirm parallel trends, see no sign of anticipatory effects or attrition, and show that differences in treatment intensity, firm characteristics, or post-invasion determinants of investments do not alter the main results.

In the discussion, I argue that the investment increase and the sectoral heterogeneity

may reflect two distinct channels through which geoeconomic risk shapes firms' decisions: a first channel in which firms reliant on Russian suppliers raise short-run investment to diversify inputs, and a second channel in which heightened uncertainty induces firms to delay investment. Financial firms, for instance, were likely more exposed to the heightened macroeconomic uncertainty that followed the outbreak of the Russo–Ukrainian war, consistent with their sharper investment retrenchment. By contrast, a diversification channel is consistent with the stronger response in utilities, where the decoupling from Russian energy inputs created an immediate need for substitution. I document that several highly exposed utilities firms report increasing investment in new oil, gas, and renewable infrastructure to reduce reliance on Russian energy imports, an adjustment supported by public interventions to provide clear policy guidance and ultimately cushion the impact of energy supply disruptions.

The paper proceeds as follows: Section 2 surveys the related literature. Section 3 details the construction of the *GRE* and presents stylised facts. Section 4 describes the empirical strategy. Section 5 shows the results. Section 6 reports robustness checks and remaining limitations. Section 7 discusses the findings. Section 8 concludes.

2 Related literature

This paper mainly relates to three literatures: (i) the emerging empirical literature on geoeconomics, (ii) the strand of work looking at how firms adjust to disruptions in international economic linkages, and (iii) the text-as-data literature that extracts measures of risk and uncertainty from textual sources.

A first strand the paper is concerned with is the growing literature on geoeconomics. This paper adopts the definition of geoeconomics proposed in Trebesch and Mohr (2025), who define geoeconomics as the study of how political rivalry, including direct conflict, shapes economic policies and outcomes, and vice versa. This broader framing is particularly relevant for the setting of this paper: in Europe, the invasion of Ukraine and the associated expansion in the use of geoeconomic instruments (e.g., sanctions and export controls) jointly altered firms' operating environments and constraints in ways that plausibly matter for investment decisions. The most prominent geoeconomic tools surveyed in the literature include sanctions,

export controls, embargoes, and tariffs. For an excellent overview of the contributions on each of these tools and more details on geoeconomics, see Trebesch and Mohr (2025). A complementary review on the literature assessing the relationship between trade and war can be found in Thoenig (2024), and in Clayton et al. (2025b) on the different conceptualisations of geoeconomic power.

Within this broader literature, the paper is most closely connected to research on the firm-level effects of geoeconomic shocks. A large body of previous work on (policy) uncertainty documents that heightened uncertainty typically depresses firms' investment and employment (see, e.g., Bloom (2009); Kang et al. (2014); Baker et al. (2016)). Geoeconomic shocks, however, may trigger additional and potentially offsetting adjustment margins. Unlike many aggregate uncertainty shocks, geoeconomic measures can directly constrain firms' transactions with specific counterparties and jurisdictions, forcing the reconfiguration of supply chains, production locations, and technology choices. Consistent with this mechanism, Alfaro and Chor (2023), studying the 2017–2018 US–China trade war, document the reallocation of supply chains toward alternative locations, including Vietnam and Mexico. Relatedly, evidence suggests that the geography of foreign direct investment has increasingly aligned with geopolitical considerations, with flows shifting toward politically closer partners (Aiyar et al. (2024); Gopinath et al. (2025)). At the firm level, qualitative evidence in Clayton et al. (2025a) indicates that firms facing geoeconomic pressure report increasing investment—including R&D—as they seek to reduce strategic dependencies. Liu et al. (2025) provide complementary empirical evidence: Chinese firms exposed to U.S. export controls increase R&D investment and patenting in relevant technologies, consistent with a substitution toward domestic capability when access to foreign inputs becomes uncertain. Finally, a related set of studies examines the firm-level effects of sanctions on the sender country. Crozet et al. (2021) show that new sanctions on Russia significantly reduced the likelihood that firms form new relationships with Russian firms, while Crozet and Hinz (2020) document that many firms not directly constrained by sanctions nevertheless curtail exports to the sanctioned destination, plausibly reflecting heightened country risk and compliance frictions. Besedeš et al. (2021) find that German firms expand their activities to non-sanctioned countries without negative effects on these firm's performance. Taken together, this literature suggests that geoeconomic

shocks can generate both a standard uncertainty channel (depressing investment) and a reconfiguration channel (raising investment through reshoring, supplier substitution, and technology upgrading), making the sign and timing of the investment response an empirical question.

A final strand relates to the use of textual sources to measure risk exposure and uncertainty. A seminal contribution is Baker et al. (2016), who construct an Economic Policy Uncertainty (EPU) index from newspaper text by tracking the co-occurrence of terms related to uncertainty and policy. Their validation exercises show that the index captures salient episodes of uncertainty and predicts subsequent declines in investment and economic activity. Subsequent work demonstrates that similar methods can be applied to alternative text sources to improve "signal-to-noise" quality or expand coverage; for example, Ahir et al. (2022) build country-level uncertainty measures using Economist Intelligence Unit reports and show that higher uncertainty is associated with significant output declines, particularly in settings with weaker institutions and tighter financial constraints. Most directly related to the present paper is the line of work extracting firm-level risk measures from earnings call transcripts, including Hassan et al. (2019), Caldara and Iacoviello (2022), Niepmann and Shen (2024), and Clayton et al. (2025a). As discussed at length in Hassan et al. (2019), these can offer a highly valuable insight into the key risks and opportunities a company is exposed to. In its first application, Hassan et al. (2019) show that earnings calls allow for the systematic identification of political risks a company is exposed to. Studying US firms, they find that the discussions on political risk are truly firm-specific, and cannot be explained by aggregate, or even sector political risks. Firms exposed to higher political risk reduce employment and investments, and increase their lobbying efforts. Earnings calls transcripts have also been used to estimate firm-level exposures to geopolitical risk. Caldara and Iacoviello (2022), who built the first index of geopolitical risk using newspapers, also proposed such an index at the firm-level by identifying discourse on geopolitics in firm's earnings calls. Firms with higher exposure to geopolitical risks are found to decrease investments and hiring in the quarters following a shock. Niepmann and Shen (2024) expand on the approach by combining this measure with bank's lending exposure by country to identify bank-level exposures to geopolitical risks. They highlight that banks reduce cross-border lending when countries experience higher geopolitical risk. Lastly,

Clayton et al. (2025a) use corporate earnings calls to analyse the geoeconomic pressure a company is facing from a foreign country. To identify instances of geoeconomic pressure, they use Large Language Model prompt engineering techniques on the transcripts. Their paper demonstrates the rise in the use of geoeconomic pressure faced by companies over the last two decades, with Chinese and Russian firms most frequently reporting being subjected to export controls and sanctions respectively, while the US, China, and the EU led in applying pressure on foreign entities. The paper also reports that, in response to the geoeconomic pressure they are facing, firms (both in the countries applying and receiving the pressure) say that they will be increasing investments, especially domestically.

Against this background, the paper makes three contributions. First, it provides a new, systematic measure of firm-level geoeconomic risk exposure, complementing existing earnings-call-based indices by combining the measurement of geopolitical risk with salience of exposure to geoeconomic instruments. Second, it brings evidence to the debate on the economic impacts of geoeconomic fragmentation by leveraging the invasion of Ukraine and the subsequent policy rupture to estimate how differential exposure to geoeconomic risk translates into investment dynamics. Third, it delves into sectoral differences and provides a discussion on how the opposing investment response of firms in the utilities and financial sectors would be consistent with the existence of geoeconomic shocks affecting firm's investment through different channels.

3 The GRE: A new measure of firm-level geoeconomic risk exposure

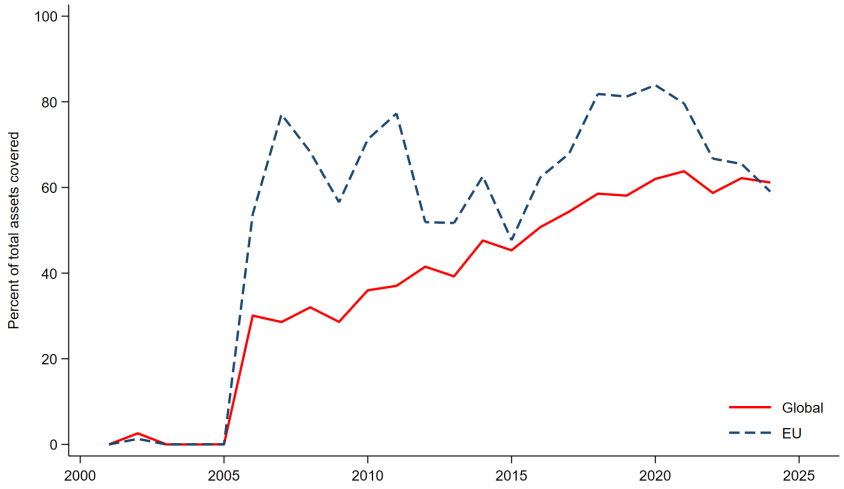
3.1 Data

The data used to build the *GRE* are the transcripts of company earnings calls. Earnings calls provide a rich source of firm-level information typically occurring every quarter. Following a presentation of the company's earnings, external analysts can ask questions in a Q&A session, on both backward- and forward-looking matters they would like to get more details on. As a result, transcripts not only reflect the firm's perspective on past and future performance, which could be biased, but also includes a demand for honest assessments of key risks and opportunities identified by external analysts. Given the analysts' material interests in the

company's performance, I assume that analysts only ask questions they truly believe are of relevance to the company's performance. This setting makes corporate earnings calls especially interesting to estimate the salience of specific risks.

The transcripts come from the NL Analytics platform which covers more than 400,000 transcripts for roughly 14,000 publicly listed firms across over 80 countries. The sample starts in 2002 and is updated on a bi-weekly basis. Examining coverage in terms of the share of total assets represented by firms with earnings calls, I find that coverage in the global sample increases steadily over time, reaching roughly 60% after 2020 (Figure 3.1). For EU firms, coverage is more volatile but remains consistently high, fluctuating between about 50% and 80% since 2005. Overall, these patterns indicate that the earnings-call sample captures a substantial fraction of aggregate corporate balance sheets, particularly in the EU.

Figure 3.1: Dataset coverage over time

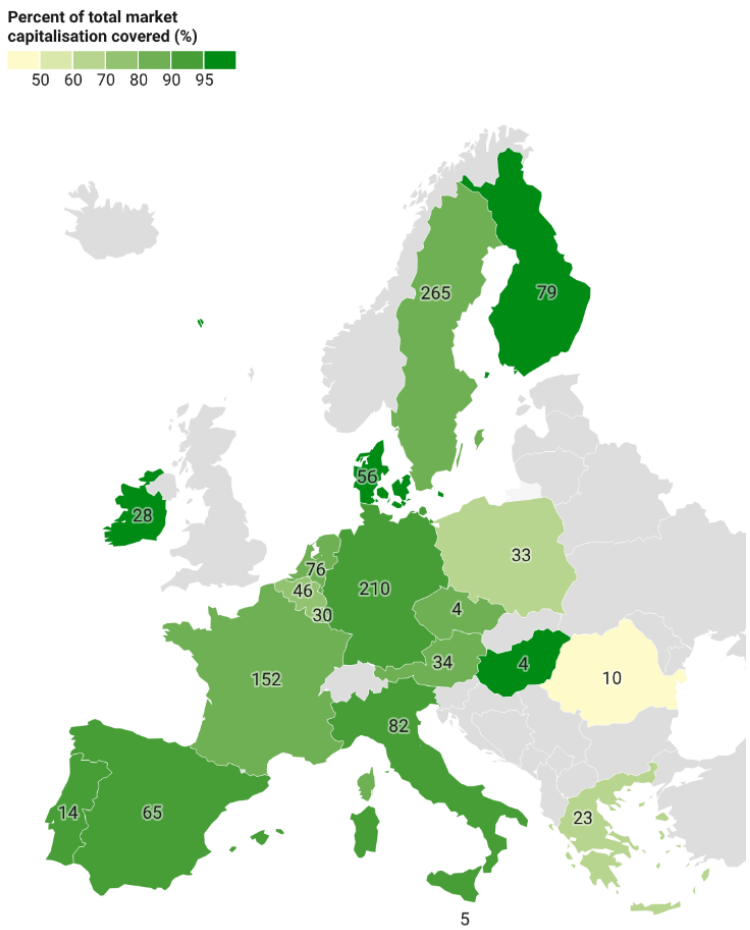


Notes: This Figure shows the share of total assets covered by firms holding an earnings call over time for the global sample and the EU sample. The data source for total assets is Compustat.

There is some important heterogeneity in the number of firms by country present in the dataset, in line with what has been documented in past literature using earnings calls (e.g. Clayton et al. (2025a)). The largest share of firms in the EU sample is headquartered either in the block's largest economies or in countries with high English proficiency, notably the Scandinavian countries. Accordingly, Sweden (20.2%), Germany (15.9%), France (11.4%), Italy (7.1%), and the Netherlands (6.6%) account for the five largest shares of firms covered.

Nonetheless, as shown in Figure 3.2, the total market capitalisation covered by these firms in 2019[1] exceeds 70% in almost all EU countries, with the exception of Malta (16.4%), Romania (40.5%), Greece (61.6%), and Poland (66.1%). This high share of covered market capitalisation supports the view that the dataset provides sufficient coverage of the largest firms within each country to extract meaningful insights from the earnings-call transcripts. Appendix A documents the data cleaning steps undertaken ahead the construction of the index construction.

Figure 3.2: Dataset coverage for EU firms in 2019



Notes: This Figure shows the number of firms by their headquarters country in the EU covered by the NL Analytics dataset and the combined percent of market capitalisation of those firms for each country in 2019. Source: NL Analytics (2026). See Table A.1 for a tabular ranking.

[1]The data on market capitalisation covered is provided by NL Analytics (2026).

It is important to note at this stage that the NL Analytics platform does not provide access to full transcripts, but run keyword queries internally. This constraint limits the application of alternative text-as-data methods—particularly LLM-based annotation schemes as in Clayton et al. (2025a)—and does not allow for systematic validation of the annotation results returned from the search query.

3.2 Methodology

To measure firm-level geoeconomic risk exposure from the earnings call transcripts, I identify the frequency of word occurrences related to geoeconomic risks in each sentence of a transcript, closely following the approaches in Hassan et al. (2019), Niepmann and Shen (2024), and Caldara and Iacoviello (2022): I define a set of keyword dictionary $\mathcal{K}$ corresponding to geopolitical developments, and geoeconomic instruments, including references to tariffs, sanctions, export controls, and embargoes. In addition, a risk dictionary $\mathcal{R}$ is employed, which identifies words related to risks, threats, and uncertainty. The dictionaries combine those deployed in Caldara and Iacoviello (2022) and Niepmann and Shen (2024) to identify geopolitical threats and acts, the one from Caldara et al. (2020) for trade-related risks, and are complemented by additional keywords for other geoeconomic instruments such as sanctions and export controls. Combining the geoeconomic and risk dictionaries helps mitigate the likelihood of capturing false positives, i.e. passages that mention geoeconomic developments irrespective of the risks they represent. The complete set of words entering the query can be found in Table A.2. Deploying the geoeconomic and risk-related dictionaries, for each sentence of an earnings call a binary measure is created by counting if the sentence simultaneously contains a geoeconomic keyword and a risk-related keyword, and aggregated over the entire transcript:

$$GRE(T) \;=\; \sum_{s \in T} \mathbb{1}\{\, \exists\, k \in \mathcal{K} : k \in s \,\} \cdot \mathbb{1}\{\, \exists\, r \in \mathcal{R} : r \in s \,\}\,,$$

with $s \in T$ the sentences of transcript T.

At the same time, geoeconomic risk may be mentioned alongside other relevant topics such as technological and market developments. Hence, since the measurement takes place at the sentence level, the index is normalised by the number of sentences N in each transcript

T to reflect the prominence given to geoeconomic risk in the discussion. This provides a transcript-level measure of GRE, with comparability across firms and over time. The GRE of any firm i is then defined as:

$$GRE_{ict} = \frac{GRE(T)_{ict}}{N(T)_{ict}},$$

with c, the country in which the firm is headquartered and t, the quarter in which the earnings call takes place.

Illustrative example. In its earnings call on 16 April 2025, a manager of ASML Holding NV noted:

> *"If AI demand continues to be strong and customers are successful in bringing on additional capacity online to support that demand, there is potential opportunity towards the upper end of our range.* ***On the other hand, there are also risks related to customers and geopolitics that could drive results towards the lower end of the range****."*

This excerpt highlights how earnings calls explicitly link firm prospects to geoeconomic risks. Indeed, Financial Times (2025) reporting documented the same day that ASML was particularly exposed to geoeconomic risks because its lithography machines risked being subjected to export controls from the U.S. administration on China.

3.3 Stylised facts

Fact 1: Geoeconomic risk salience rises sharply after major geoeconomic events, and is conditioned by geographic exposure. Figure 3.3 plots the standard deviation of the aggregate geoeconomic risk exposure for firms headquartered in the European Union and in the full global sample.[2] Looking at the development of the GRE over time, two

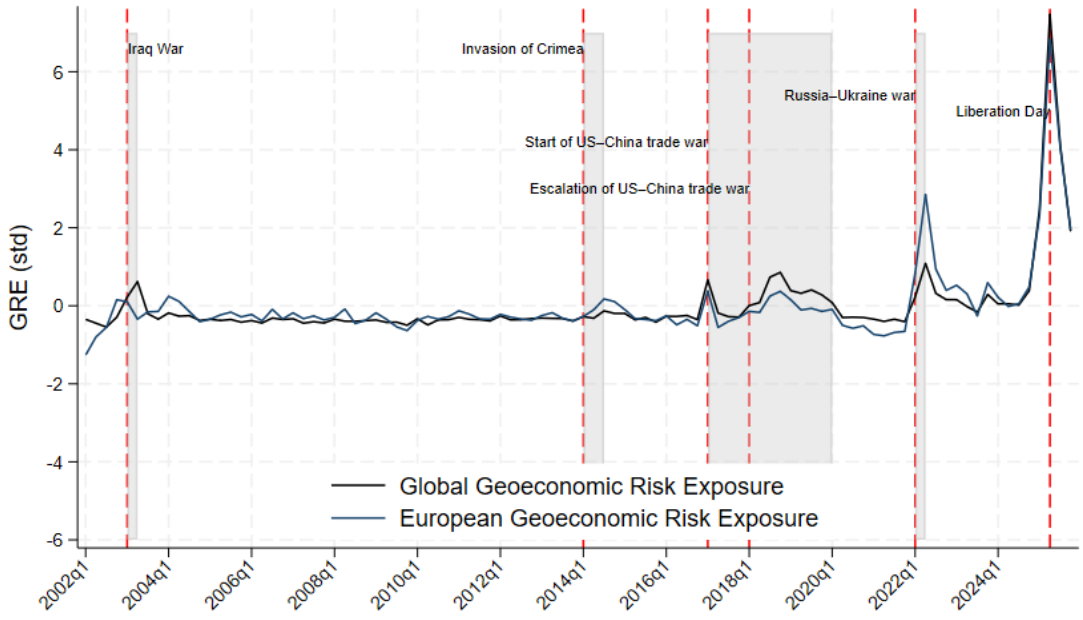
[2]The aggregate indices sum firm-level exposure over firms with at least one call in quarter t and are standardized to mean zero and unit variance over the sample period to facilitate comparability over time. Formally,

$$GRE_t^{EU} = \sum_{c \in \mathcal{C}_{EU}} \sum_{i=1}^{F_{ct}} GRE_{ict}, \qquad GRE_t^{W} = \sum_{c \in \mathcal{C}_{W}} \sum_{i=1}^{F_{ct}} GRE_{ict},$$

regularities stand out. First, geoeconomic risk discussions are comparatively absent prior to the mid-2010s, with the exception of the Iraq war, before increasing on five instances thereafter coinciding with the annexation of Crimea, the US-China trade war, the Russian invasion of Ukraine, and Liberation Day. This increase in salience of geoeconomic risks at the corporate level is consistent with greater presence of geoeconomic pressure identified in Clayton et al. (2025a). Second, with exception of the escalation of the US-China trade war in 2018, the spikes in the *GRE* are typically short-lived.

Comparing the EU and global indices, we can see that they co-move around major events but differ in amplitude across episodes: the global index rises more strongly around episodes centred on US-China trade tensions, whereas the European index displays particularly pronounced spikes around Russia-related shocks, including the annexation of Crimea, and especially after the full-scale invasion of Ukraine.

Figure 3.3: Geoeconomic Risk Exposure: EU vs World



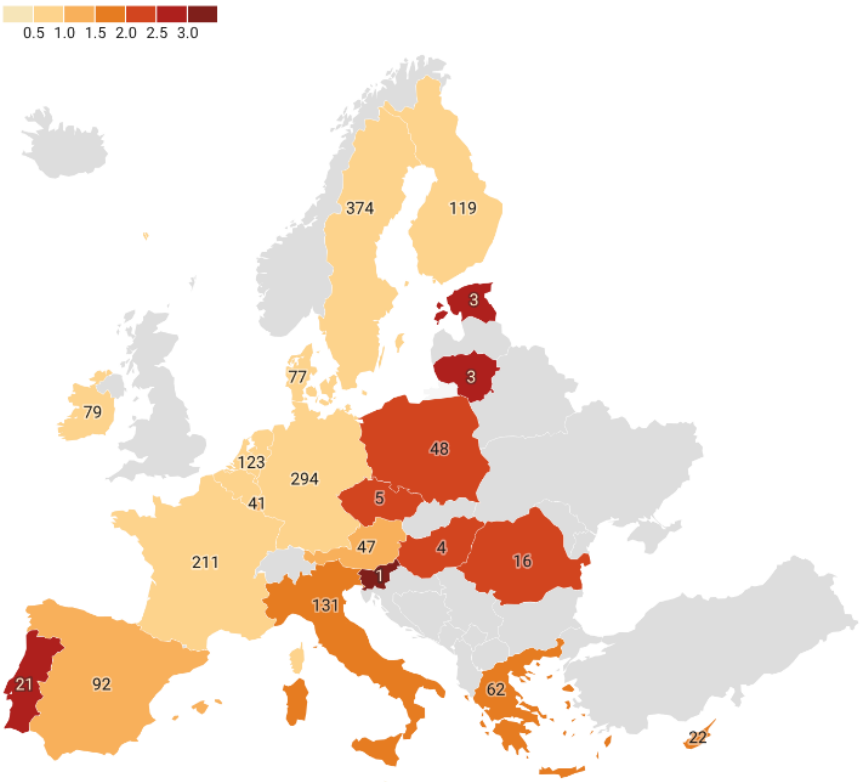
Notes: This Figure shows the standardised *GRE* for firms headquartered anywhere in the world (black line), and for those headquartered in the European Union (blue line). Dotted red lines indicate the quarter of an event labelled next to it. Grey areas show the period of heightened uncertainty that followed the event. For the full search query see Table A.2.

Looking more closely at the geographic exposure in the EU, I find that on aggregate over

where F_{ct} denotes the number of distinct firms headquartered in country c that hold at least one call in quarter t.

the entire time period firms from Eastern European countries are relatively more exposed compared to firms from the rest of the EU (Figure 3.4). Poland and Romania display average exposure levels that are around twice those observed in France or Spain, while the Baltic states record some of the highest values in the sample. Italy and Greece also exhibit higher average exposure than Germany and the Nordic countries. By contrast, France, Germany, Sweden, and Finland fall in the lower-to-middle range of the distribution. The map also indicates that exposure tends to be higher in countries geographically closer to Russia. Finally, cross-country comparisons should be interpreted in light of differences in sample coverage: some countries are represented by many more firms than others, which may affect the precision and representativeness of the country averages.

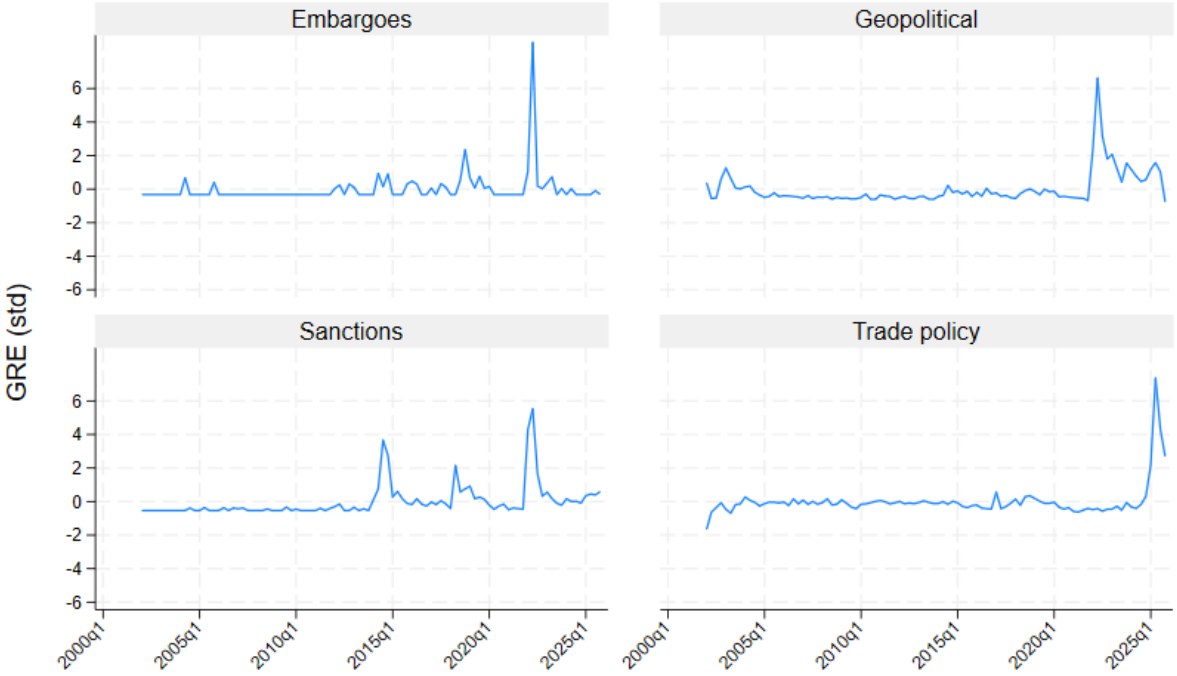
Figure 3.4: Geoeconomic Risk Exposure by country



Notes: This Figure show the average *GRE* by country, and the number of firms per country. The colour scheme indicates the average number of geoeconomic risk mentions per 1,000 sentences over the entire time period. See Table A.3 for a tabular ranking, and Table A.2 for the full search query.

Fact 2: Periods of elevated geoeconomic risk reflect a broadening in the mix of instruments. Disaggregating the European *GRE* into its sub-components shows that the composition of geoeconomic risk is episode-specific (Figure 3.5). In some periods, individual instruments become particularly salient. For example, sanctions feature prominently around Russia's annexation of Crimea, while trade-policy language intensifies during phases of tariff escalation. However, the most pronounced spike for the geopolitical sub-component around Russia's full-scale invasion of Ukraine is characterized by the simultaneous rise of other components. This pattern is consistent with European firms facing not just facing disruptions due to the effects of the war, but through a bundled mix of geoeconomic instrument which jointly shape perceived risks and adjustment needs. As a result, a broad-based index is informative precisely when risks are highest: narrower measures focusing only on geopolitical developments or on sanctions alone would miss substantial variation in the policy mix captured in earnings-call discussions.

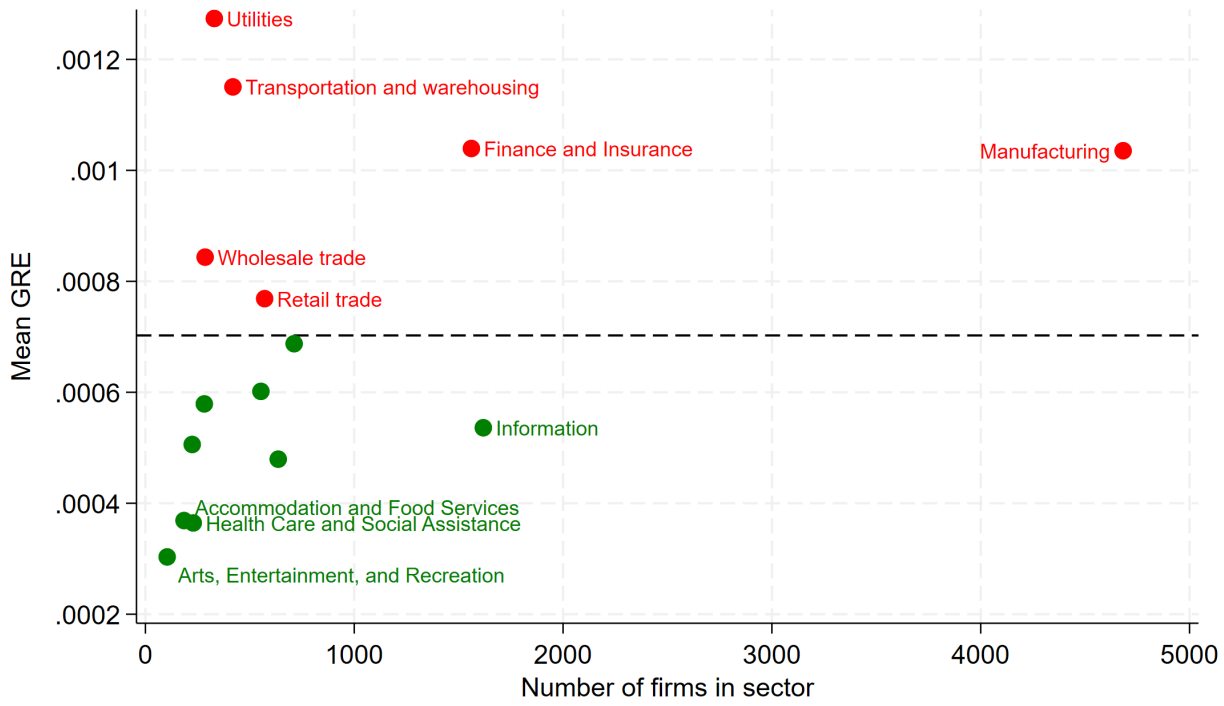
Figure 3.5: European Geoeconomic Risk Exposure: Sub-components



Notes: This Figure shows the standard deviation of the *GRE* split by its sub-components. For the query terms included for each sub-component see Table A.2.

Fact 3: Sectoral exposure to geoeconomic risks is highly heterogeneous. Aggregating firm-level exposure to the sector level[3] reveals substantial heterogeneity in average geoeconomic risk salience (Figure 3.6). Firms in the Utilities sector exhibit the highest average exposure over the sample, closely followed by Transportation and warehousing, Finance and Insurance, and Manufacturing. This is consistent with the greater exposure of these sectors to disruptions in global supply chains. By contrast, more domestically oriented sectors—such as the Arts, Entertainment, and Recreation sector—display geoeconomic risk exposure that is twice as low.

Figure 3.6: Geoeconomic Risk Exposure by sector



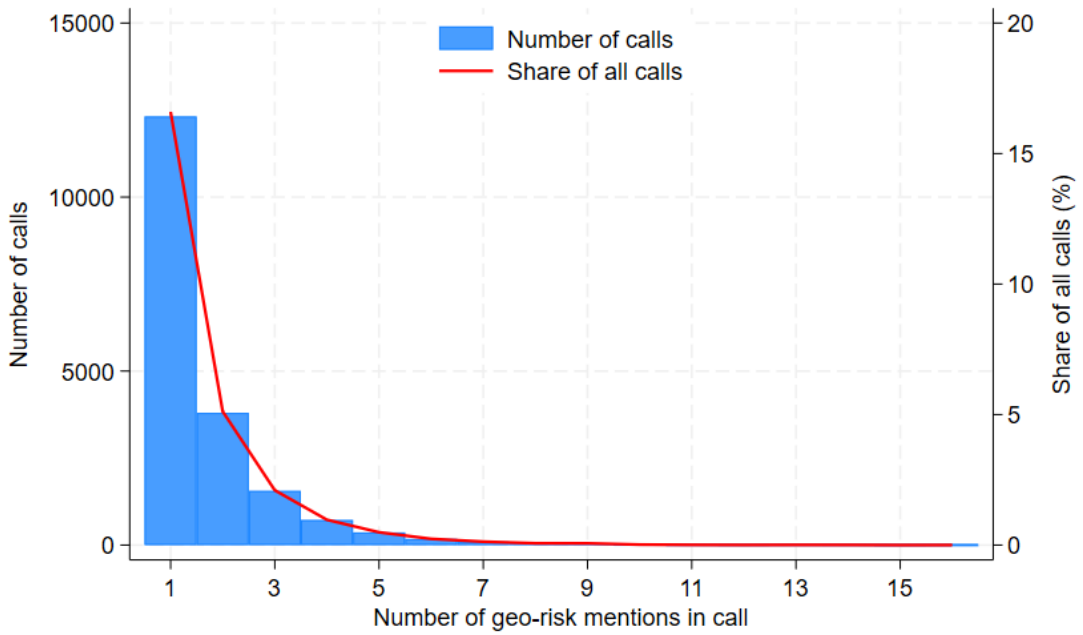
Notes: This Figure shows the average *GRE* for all sectors with over 100 firms, and the number of firms within that sector. Red and green dots indicate whether a sector's average is above or below the mean *GRE* respectively. Sectors are defined by their NAICS classification in Compustat. For the full search query see Table A.2.

Fact 4: Geoeconomic risks are mentioned at least once in a quarter of all earnings calls, but repeated mentions in one call are rare. Geoeconomic risks are mentioned

[3]Defined by the North American Industry Classification System (NAICS) codes taken from Compustat.

at least once in 25.8% of European earnings calls (Figure 3.7). The distribution of mentions is strongly left-skewed: among calls that reference geoeconomic risks, most do so only once, while calls with intensive discussion (e.g., five or more mentions) account for just over 1% of all calls. This pattern is consistent with geoeconomic risks being a recurring but typically non-dominant topic in corporate communications, punctuated by a small number of calls in which these issues feature prominently.

Figure 3.7: Number of sentences mentioning geoeconomic risk per call



Notes: This Figure plots the number of earnings calls by sum of sentences mentioning geoeconomic risks in that call (left-axis) and the share of all calls that this number of calls correspond to (right-axis). For the full search query see Table A.2.

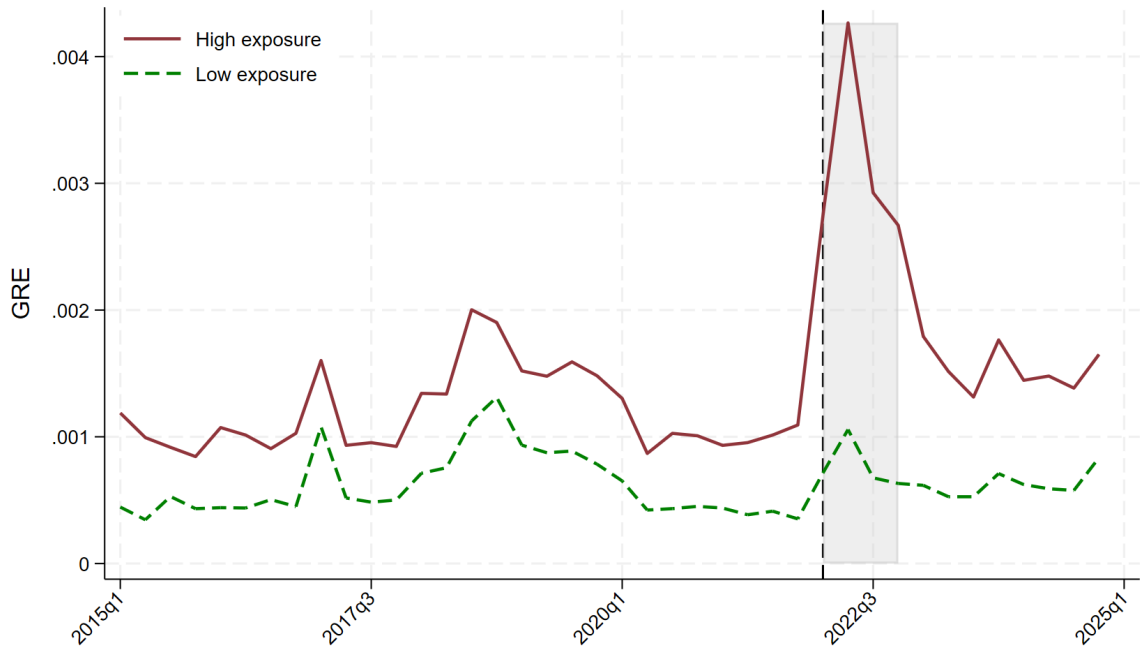
4 Empirical Strategy

To assess the causal effect of geoeconomic risk exposure on firm performance, the empirical strategy follows a difference-in-differences (DiD) design, using the Russian invasion of Ukraine in 2022 as a quasi-exogenous shock to the perceived salience of geoeconomic risks. As depicted in Figure 4.1 the discussions in earnings calls showed no anticipation of a growing risk before the full-scale invasion in Q1 2022. Treatment for the DiD is based on whether firms' average *GRE* (i.e. the average mentions of geoeconomic risks in an earnings call) is above or below the

median for 2022, the first year of the Russo-Ukrainian war. I look at the entire year following the invasion to define the treatment taking into account the fact that some companies only hold bi-annual earnings calls and that the discussions on geoeconomic risks continued after Q1 2022, possibly due to the discussions shifting to the effects of embargoes and sanctions. Formally, the treatment indicator is defined as

$$D_i = \mathbb{1}\left(\frac{1}{T_{2022}} \sum_{t \in 2022} GRE_i > \text{median}\left[\frac{1}{T_{2022}} \sum_{t \in 2022} GRE_i\right]\right).$$

Figure 4.1: Mean GRE over time for treatment and control groups



Notes: This Figure depicts the mean *GRE* for firms which are in the treatment group (High exposure) and for those which are in the control group (Low exposure). The grey area shows the period of heightened uncertainty that followed the invasion.

Plotting the average *GRE* over time for these two groups shows that the spike following the beginning of the war in Ukraine is indeed much larger for the treatment group than for the control group (Figure 4.1). Another important insight is that the *GRE* remains more elevated for longer after the start of the war, possibly reflecting that discussions shifted from purely geopolitical risks to the effects of sanctions and export controls that followed the

invasion. This reinforces the importance of including these terms in our analysis to have a more complete picture of firm's exposure to geoeconomic risks, rather than just those of war. Moving now to the impact of this exposure on firm-level investments, I estimate the following specification:

$$Y_{it} = \alpha + \beta\, \mathrm{D}_i \times \text{Post-invasion}_t + \mathbf{X}'_{it}\boldsymbol{\gamma} + \delta_t + \lambda_y + \sigma_i + \varepsilon_{it}, \tag{4.1}$$

where Y_{it} denotes the log of real capital expenditures for firm i in quarter t. D_i is the dummy indicator for splitting firms into high / low geoeconomic exposure [4]. Post-invasion$_t$ is a dummy that takes the value of 0 before Q1 2022 and 1 otherwise. δ_t denote quarter fixed effects, λ_y denote year fixed effects, and σ_i are firm fixed effects. $\mathbf{X}_{it}$ is a vector of time-varying firm-level control variables. α is the constant and standard errors are clustered at the firm level.

The coefficient of interest is β, which captures the average treatment effect on the treated (ATT), i.e. for firms with higher-than-median geoeconomic risk exposure in the first year of the war. Identification relies on the standard parallel trends assumption in this fixed-effects setting: absent the invasion, treated and control firms would have exhibited similar investment dynamics over time. Plotting the capital expenditure for the two groups over time gives preliminary confidence in this assumption: the investments of the two groups largely co-move until 2022 and are of very similar magnitude. This co-movement is important in a DiD setting to ensure there are no differences in behaviour prior to the shock, and is discussed in greater detail in the robustness section. After the start of the war, the capital expenditures of the two groups diverge. Firms in the highly exposed groups continue increasing their capital expenditures while firms in the low exposure group cut investments.

The choice of fixed effects is central to the estimation of effects. Time fixed effects, δ_t, absorb all shocks that are common to firms in a given quarter, such as aggregate demand conditions, or monetary and financial conditions. Without δ_t, any post-2022 macro shock that mechanically affects investment could be erroneously attributed to the treatment indicator. In addition, year fixed effects, λ_y, are also added to the estimation to ensure that the effects

[4] By construction firms which do not have an earnings call in 2022 are excluded from the analysis. Risks of attrition are addressed further down below.

Figure 4.2: Parallel trends assumption

Notes: This Figure depicts the mean log real capital expenditure for firms which are in the treatment group (High exposure) and for those which are in the control group (Low exposure).

of variables for which there is only annual data, are also appropriately controlled for. Finally, firm fixed effects, σ_i, remove time-invariant differences across firms, including, for instance, persistent heterogeneity in size, revenues, and liquidity.

I report summary statistics for firms split by high/low exposure and the complete set of summary statistics in Tables B.1 and B.2 respectively. The final sample comprises 316 high-exposure and 330 low-exposure firms. The results from the two-sided t-tests between high and low exposure firms, with standard errors clustered at the firm level, motivate controlling for observable firm characteristics in the regression analysis. While differences across groups are modest in economic magnitude, high-exposure firms are, on average, slightly more equity-financed, appear marginally smaller and register somewhat lower sales. Accordingly, the inclusion of time-varying controls $\mathbf{X}_{it}$ complements the fixed effects by accounting for observable changes in firms' balance sheets over time. Specifically, $\mathbf{X}_{it}$ includes equity-to-assets, cash flow, total assets, sales, cost of goods sold, debt, property, plant and equipment, and interest expenses. I use logs for all level variables (except for ratios) and lag

controls by one quarter to reduce concerns about reverse causality, that current investment mechanically may affect contemporaneous balance-sheet outcomes. All nominal variables are deflated to have their real values. See Appendix B for the full description of the data cleaning steps.

5 Results

5.1 Main results

Table 5.1 reports the main results from the baseline specification reported in equation (4.1). Once the full set of fixed effects and firm-level controls are accounted for, I find that firms which were more exposed to geoeconomic risks in the first year of the Russo-Ukrainian War increased investments by roughly 12% after 2022. In terms of magnitude, this increase in investments represents about double the average annual investment growth rate for these firms[5].

In column (1), which includes firm fixed effects only, the coefficient on $D_i \times$ post-invasion is positive but small (0.037) and statistically insignificant. Once quarter fixed effects are added in column (2), the estimate rises to 0.133 and becomes statistically significant at the 5% level. The coefficient remains similar when the specification is further tightened. In column (3), adding year fixed effects and a richer set of lagged firm-level controls yields an estimate of 0.126, significant at the 1% level. Column (4) additionally includes controls for inflation and the central bank policy rate; the coefficient remains positive and statistically significant (0.121 at the 5% level). The stability of the estimate across columns (2)–(4) suggests that the baseline result is not driven by changes in firms' balance-sheet characteristics or financing conditions, but reflects a differential post-invasion adjustment associated with geoeconomic exposure.

The control coefficients are broadly consistent with standard investment determinants. Firm size and activity are positively associated with investment, as reflected in the highly significant coefficients on log assets and log sales. Higher leverage and interest expenses are associated with lower investment, consistent with tighter financing constraints. Equity-to-

[5]The average annual investment growth rate for firms included in the estimation is 6.9% (sd 0.16).

Table 5.1: Main results

	(1)	(2)	(3)	(4)
$D_i \times$ post-invasion	0.037	0.133**	0.126***	0.121**
	(0.042)	(0.053)	(0.048)	(0.048)
Equity-to-assets	0.269**	0.242**	0.225	0.229
	(0.124)	(0.121)	(0.145)	(0.145)
Cash flow	-0.304*	-0.320*	-0.038	-0.040
	(0.165)	(0.167)	(0.125)	(0.124)
Log assets	0.802***	0.824***	0.306***	0.308***
	(0.042)	(0.043)	(0.087)	(0.087)
Log sales			0.293***	0.291***
			(0.059)	(0.059)
Log COGS			0.009	0.011
			(0.035)	(0.034)
Log debt			-0.036***	-0.035***
			(0.009)	(0.009)
Log PPE			0.371***	0.368***
			(0.067)	(0.067)
Log interest exp.			-0.049***	-0.049***
			(0.012)	(0.012)
Inflation				0.008
				(0.011)
CB policy rate				0.015
				(0.019)
Observations	25,448	25,448	24,558	24,531
Firm FE	Yes	Yes	Yes	Yes
Quarter FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes

Notes: This Table shows the effects of interacting the treatment dummy D_i (higher-than-median GRE_i in 2022) with a $post - invasion_t$ dummy on log real capital expenditures. Included controls are in columns (1)-(2): equity-to-assets, cash flow, and log assets; column (3) adds log sales, log cost of goods sold, log debt, log plant, property, and equipment (PPE), log interest expenditures; and column (4) adds the country's inflation rate and central bank policy rate. Column (4) is the baseline corresponding to equation (4.1). All firm-level controls are lagged. Robust standard errors clustered at the firm level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

assets and cash flow become statistically insignificant once additional controls are added, indicating that the main effect is not primarily operating through equity financing or short-run liquidity. The other controls (COGS, inflation, and central bank policy rate) are statistically insignificant throughout the estimations.

5.2 Sectoral differences

The heterogeneity in sectoral exposure to geoeconomic risks (Figure 3.6) motivates examining whether investment responses to the Russo–Ukrainian war differ by sector. While the baseline specification absorbs time-invariant sectoral differences,[6] treatment effects may still vary across sectors.

To estimate sectoral differences I adapt equation (4.1), and interact the treatment interaction ($D_i \times$ Post-invasion) with sector dummies:

$$Y_{it} = \alpha + \beta\, \mathrm{D}_i \times \text{Post-invasion}_t + \mathrm{D}_i \times \text{Post-invasion}_t \times \text{Sector}_{is} + \mathbf{X}'_{it}\boldsymbol{\gamma} + \delta_t + \lambda_y + \sigma_i + \varepsilon_{it}, \quad (5.1)$$

where Y_{it} denotes the log of real capital expenditures for firm i in quarter t. D_i is the dummy indicator for splitting firms into high / low geoeconomic exposure. Post-invasion$_t$ is a dummy that takes the value of 0 before Q1 2022 and 1 otherwise. Sector$_{is}$ is a dummy equal to 1 if firm i belongs to sector s (e.g. manufacturing, finance, or utilities). δ_t denote quarter fixed effects, λ_y denote year fixed effects, and σ_i are firm fixed effects. $\mathbf{X}_{it}$ is a vector of time-varying firm-level control variables. α is the constant and standard errors are clustered at the firm level.

The results presented in Table 5.2 reveal that there is a strong heterogeneity in investment response for some sectors. Column (1) presents the baseline specification. In column (2), the triple interaction with manufacturing firms is not statistically different from that for non-manufacturing firms. Column (3) indicates that financial firms reduce investment by more than 45% relative to non-financial firms; the effect is robust at the 1% level. Column (4) shows that utilities firms increase investment by over 55% relative to the rest, again significant at the 1% level. Column (5) excludes financial firms from the baseline sample; the

[6] With firm fixed effects, sector fixed effects are also absorbed.

coefficient remains similar in magnitude and significant at the 1% level. Column (6) excludes utilities firms; the baseline effect remains positive and significant at the 10% level, though it is 4 percentage points smaller, implying 8.8% higher investment for firms with high GRE. Column (7), which excludes both financial and utilities firms, yields a similar estimate: a coefficient of 0.09, significant at the 10% level.

Table 5.2: Sectoral differences

	Excluded sectors						
	(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*
	None	*None*	*None*	*None*	*Fin.*	*Ut.*	*Fin. & Ut.*
$D_i \times$ Post	0.121**	0.104*	0.124**	0.077	0.129***	0.088*	0.091*
	(0.048)	(0.056)	(0.048)	(0.048)	(0.049)	(0.050)	(0.050)
$D_i \times$ Post $\times$ Manuf		0.061					
		(0.070)					
$D_i \times$ Post $\times$ Finance			-0.595***				
			(0.050)				
$D_i \times$ Post $\times$ Utilities				0.440***			
				(0.117)			
Firm controls	✓	✓	✓	✓	✓	✓	✓
Observations	24,531	24,531	24,531	24,531	24,440	22,794	22,676
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This Table shows the results of equation (5.1). The sector interaction dummy is for Manufacturing (2), Finance (3), and Utilities (4) firms respectively. Columns (5)-(7) exclude from the sample firms in the financial sector, utilities sector, and both respectively. All specifications include for firm-level controls equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and for country-level controls inflation and the central bank policy rate. All firm controls are lagged. Robust standard errors clustered at the firm level in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.3 Effect on other firm-level variables

The effect of geoeconomic exposure on firms is found to be exclusively relevant to their investments. I estimate equation (4.1) but replace the dependent variable (log capital expenditures) with log assets, log interest expenditures, equity-to-assets, log sales, log debt,

and log cost of goods sold in columns (1)-(6) respectively. I find that above-median *GRE* in 2022 (the treatment) does not have a statistically significant effect on any firms variables, except for firm's cost of goods sold significant at the 10% level (Table 5.3).

Table 5.3: Effect on other firm variables

	Dependent variable					
	(1) Assets	*(2) Int. exp*	*(3) Equity-to-assets*	*(4) Sales*	*(5) Debt*	*(6) COGS*
$D_i \times$ Post-invasion	-0.016	-0.056	-0.003	0.026	0.024	0.051*
	(0.021)	(0.056)	(0.010)	(0.024)	(0.073)	(0.030)
Firm controls	✓	✓	✓	✓	✓	✓
Observations	24,570	24,566	24,984	24,569	24,588	24,556
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This Table the results of equation (4.1), replacing log real capital expenditures as the dependent variable by (1) log total assets, (2) log interest payments, (3) equity-to assets, (4) log sales, (5) log debt, (6) log cost of goods sold. All specifications include for firm-level controls equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and for country-level controls inflation and the central bank policy rate. All firm controls are lagged. Robust standard errors clustered at the firm level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

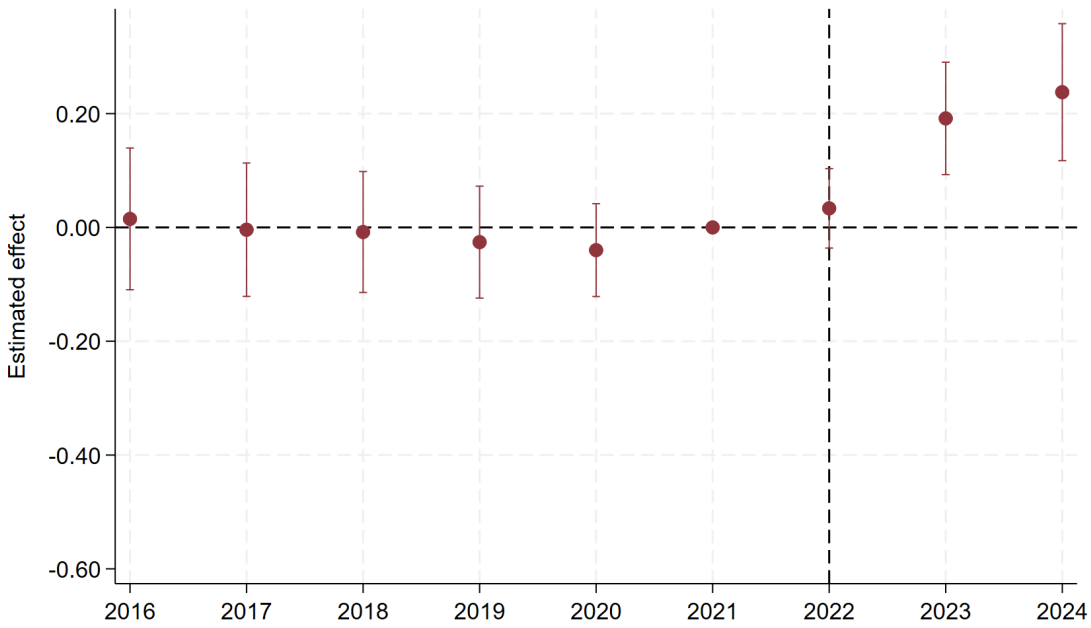
5.4 Dynamic treatment effects

To investigate the dynamic treatment effect, i.e. how higher exposure to geoeoconomic risks in the first year of the Russo-Ukrainian war affects investments for each year following the invasion, I adapt the main specification to follow an event-study design:

$$Y_{it} = \alpha + \sum_{k=2016}^{2024} \beta_k \left(\mathrm{D}_i \times k\right) + \mathbf{X}'_{it}\boldsymbol{\gamma} + \delta_t + \lambda_y + \sigma_i + \varepsilon_{it}. \tag{5.2}$$

where Y_{it} denotes the log of real capital expenditures for firm i in quarter t. D_i is the dummy indicator splitting firms into high / low geoeconomic exposure groups. k is the event time. δ_t denote quarter fixed effects, λ_y denote year fixed effects, and σ_i are firm fixed effects. $\mathbf{X}_{it}$ is a vector of time-varying firm-level control variables. α is the constant and standard errors are clustered at the firm level.

Figure 5.1: Estimated effect over time



Notes: This Figure plots β_k of equation (5.2), i.e. the coefficient of the interaction term between the high geoeconomic exposure dummy and the event time. The regression includes for firm-level controls equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and for country-level controls inflation and the central bank policy rate. All firm-level controls are lagged. The estimation includes firm, quarter, and year fixed-effects. Robust standard errors are clustered at the firm-level. Whiskers are the 95 percent confidence intervals of the coefficient.

It appears that the effect becomes significant one year after the start of the war (Figure 5.1). The non-significance of the interaction term prior to the war is crucial to address concerns about pre-existing differences between firms in the two groups. In the two years that follow start of the war, the coefficient becomes statistically significant and jumps to around 0.2, implying a 20% increase in investments per year for highly exposed firms relative to the control group. This increase would represent an even larger effect in terms of magnitude than the one found in the baseline regression, amounting to over three times the average annual growth rate in investments for the firms in the sample.

5.5 Effects outside of the EU

Expanding the analysis beyond firms headquartered in the EU, I find that the investment response to geoeconomic risk exposure following the invasion is largely concentrated among EU firms, and does not generalise in a statistically discernible way to firms headquartered outside the EU. Table 5.4 estimates the baseline specification across different geographic samples, with the high-GRE treatment indicator constructed separately within each sample. Column (1) sets the baseline for EU-only firms previously reported. Expanding the sample to include all firms in Column (2) attenuates the estimate substantially: the coefficient falls to 0.045 and but remains significant at the 10% level. Finally, restricting the sample to non-EU headquartered firms in Column (3) yields a small and statistically insignificant estimate (0.015).

Table 5.4: Comparison to non-European firms

	(1)	(2)	(3)
$D_i \times$ post-invasion (EU only)	0.121^{**}		
	(0.048)		
$D_i \times$ post-invasion (All)		0.045^{*}	
		(0.026)	
$D_i \times$ post-invasion (Non-EU)			0.015
			(0.030)
Firm controls	✓	✓	✓
Observations	24,558	97,806	73,248
Firm FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

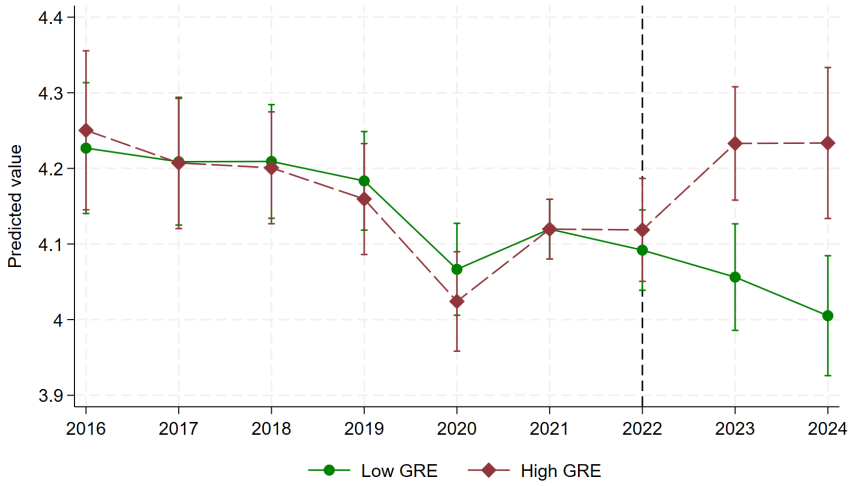
Notes: This Table shows the main estimation from equation (4.1), applied to the sample of firms headquartered in the EU (1), anywhere in the world (2), and excluding firms from the EU (3). All specifications include for firm-level controls equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and for country-level controls inflation and the central bank policy rate. All firm controls are lagged. The treatment variable D_i is calculated separately for the three regressions. Robust standard errors clustered at the firm level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6 Robustness

6.1 Parallel trends

A first robustness concern relates to the parallel trends of the high- and low-exposure firms. An objection to the descriptive comparison of capital expenditures shown in Figure 4.2 could be that these are based on raw group averages and do not condition on the firm fixed effects and control variables included in the baseline DiD specification. To assess whether the parallel-trends assumption also holds within the regression framework, I compute and plot the implied predicted investments for high- and low-GRE firms over time using equation (5.2) from the event study. The estimated pre-2022 interaction terms are statistically indistinguishable from zero, supporting parallel trends once firm heterogeneity is accounted for. After 2022, predicted investments diverge across the two groups, with statistically meaningful differences emerging from 2023 onwards, consistent with the baseline results.

Figure 6.1: Predicted investments



Notes: The Figure reports the predicted yearly log real capital expenditures for firms in the treatment (High GRE) and control (Low GRE) groups from estimating equation (5.2). The regression includes for firm-level controls equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and for country-level controls inflation and the central bank policy rate. All firm-level controls are lagged. The estimation includes firm, quarter, and year fixed-effects. Whiskers indicate 95% confidence intervals based on standard errors clustered at the firm level.

6.2 Anticipation effects

Another key issue to address in differences-in-differences setups is a possible anticipatory effect from firms with higher exposure. To address this issue, I run placebo tests by using my dummy variable for classifying firms into the high or low exposure group, and interact it with a dummy in 2014q1, 2018q1, and 2020q1, finding no evidence of anticipatory effects.

Table 6.1: Placebo tests

	(1)	(2)	(3)	(4)
D_i × Post-2014	-0.064			
	(0.054)			
D_i × Post-2018		-0.007		
		(0.050)		
D_i × Post-2020			0.041	
			(0.050)	
D_i × post-invasion				0.121^{**}
Firm controls	✓	✓	✓	✓
Observations	24,558	24,558	24,558	24,558
Firm FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This Table shows the estimating equation 4.1, but interacting the treatment dummy D_i with the quarter dummies 2014q1 (1) , 2018q1 (2), and 2020q1 (3) respectively. The regression includes for firm-level controls equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and for country-level controls inflation and the central bank policy rate. All firm-level controls are lagged. The estimation includes firm, quarter, and year fixed-effects. Robust standard errors clustered at the firm level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6.1 presents the series of placebo tests. Columns (1)–(3) report placebo estimates for post-2014, post-2018, and post-2020, respectively. Across all three specifications, the coefficients on the treatment interaction are statistically insignificant, indicating no systematic differential investment response of highly exposed firms in these earlier periods. Taken together, the placebo tests provide compelling evidence that the positive investment response of highly geoeconomically exposed firms is specific to the post-2022 period and does not reflect spurious correlations, earlier crises, or unobserved differential trends.

6.3 Risk of attrition and survivorship bias

A bias in the results could also arise if highly exposed firms stop holding earning calls. This could be either because they believe that discussing the exposure would be negatively seen by investors, or because the impact is so large that the firms simply go bankrupt. This selection into survivorship could bring an important bias to the results as these firms would be more likely to reduce investments. However, looking at the share of firms that remains in the dataset across years reduces this concern. Among firms that held an earnings call in 2020, 92% also held a call in 2022, the year of the invasion. This retention rate is close to the corresponding share for the pre-period: just over 95% of firms that held a call in 2018 also held one in 2020. These similar rates attenuate concerns that the results are driven by differential sample attrition. Figure C.1 illustrates this pattern by plotting, for the cohort of firms with an earnings call in 2018, the share that continues to hold calls in subsequent years. The series declines gradually, as expected given firm entry and exit, but the slope is essentially unchanged around 2022 and does not steepen thereafter. This stability suggests no abnormal drop in earnings-call activity following the start of the war.

6.4 Differences in exposure intensity

The baseline estimation uses a binary treatment definition to classify firms in groups of high- and low geoeconomic risk exposure. Yet, even among treated firms some firms may discuss geoeconomic risks substantially more than others. This variation in geoeconomic risk exposure, generally called differences in treatment intensity in DiD settings, could also lead to biased estimators. To account for this, I re-estimate the baseline specification using a continuous measure of exposure: the firm's *GRE* expressed in standard-deviation units, interacted with the post-invasion dummy. The positive investment response among more exposed firms continues to hold in this specification. Specifically, a one-standard-deviation increase in *GRE* is associated with a 4.4% increase in investment in the post-invasion period (Table 6.2 column (2)). Since *GRE* reaches values of roughly three standard deviations at its peak around the invasion period (see Figure 3.3), the implied magnitude is broadly consistent with the main results: $3 \times 4.4\% \approx 13\%$, close to the headline 12% increase.

Table 6.2: Main results with standard deviation of GRE

	(1)	(2)
GRE sd. × post-invasion	0.036*	0.044**
	(0.021)	(0.018)
Firm controls	✓	✓
Observations	40,335	38,764
Firm FE	Yes	Yes
Quarter FE	No	Yes
Year FE	No	Yes

Notes: This Table shows the results from estimating equation 4.1, but using the standard deviation of the *GRE* instead of the treatment dummy D_i. The regression includes for firm-level controls equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and for country-level controls inflation and the central bank policy rate. All firm-level controls are lagged. The estimation includes firm fixed effects for column (1) and firm, quarter, and year fixed-effects for column (2). Robust standard errors clustered at the firm level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.5 Post-invasion shifts in controls

Another concern may be that the difference in relevance of the control variables have a different effect on investments post-invasion, which may affect the results. I address this concern in column (2) of Table C.1 by interacting all firm- and country-level controls with the post-invasion dummy. The effect of high geoeconomic exposure remains stable in magnitude and statistically significant at the 5% level. The interacted controls also behave in intuitive ways. Higher leverage is associated with a more negative post-2022 investment response, consistent with tighter financing conditions disproportionately constraining more indebted firms. Firms with a larger pre-existing capital stock (PPE) exhibit a stronger post-2022 investment response, while the interaction on cash flow becomes more positive, consistent with internal funds gaining importance when external finance tightens. Overall, allowing for heterogeneous post-2022 effects of firm-level characteristics does not attenuate the estimated effect of geoeconomic exposure, supporting the interpretation that the baseline result is not driven by differential balance-sheet dynamics.

6.6 Firm differences

A last concern may be that there are important firm differences that remain even after including firm-level fixed effects and controls. To mitigate this risk, I re-estimate my main results after using a Propensity Score Matching technique. I weigh observations according to pre-invasion firm characteristics, and trim the dataset to remove the 5% of firms for which the pscore obatined by running the matching procedure is the lowest. See Figure C.2 for the kernel density graph obtained after the matching procedure. I find that the main results on treatment remain very similar in magnitude and are highly robust, with statistical significance at the 5% level.

Table 6.3: Results with Propensity Score Matching

	(1)	(2)
D_i × post-invasion	0.117^{**}	0.107^{**}
	(0.052)	(0.050)
Firm controls	✓	✓
Firm controls × Post-2022	✗	✓
Observations	22,978	22,951
Firm FE	Yes	Yes
Quarter FE	Yes	Yes
Year FE	Yes	Yes

Notes: This table reports results from a propensity score matching exercise. All columns apply the weights obtained from propensity score matching on all control variables using the nearest-neighbour approach, and imposing common support by trimming 5% of the treatment observations at which the pscore density of the control observations is the lowest (see Leuven and Sianesi (2018) for further details on the matching procedure). The regression includes for firm-level controls equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and for country-level controls inflation and the central bank policy rate. Robust standard errors clustered at the firm level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7 Discussion

7.1 Possible mechanisms: Two channels at play?

What mechanisms could rationalise the estimated investment response? One explanation for the baseline 12% increase in investment among high-*GRE* firms after Russia's invasion of Ukraine may be the existence of a diversification channel, in line with the literature on the reconfiguration of global value chains and investment flows (see discussion of Aiyar et al. (2024); Gopinath et al. (2025); Alfaro and Chor (2023) in Section 2). Firms with high exposure were more likely than those with low exposure to have to reconfigure suppliers, logistics, and input sourcing in response to increased country risk in Russia and Ukraine, as well as sanctions and related trade and financial restrictions. Such reconfiguration can require upfront capital expenditure to comply with sanctions, and ultimately reduce exposure to future disruptions. Consistent with this possible mechanism, Clayton et al. (2025a) document that firms discussing the invasion of Ukraine frequently report plans to increase investments.

The comparatively stronger investment increase of firms in the utilities sector also lends more confidence in the hypothesis of a diversification channel. Firms in the utilities sector, and chiefly energy firms among them, were heavily dependent on Russian energy imports and were faced with its loss through the EU's embargoes on coal and oil in August/October 2022, as well as the stop of gas flows from Russia (McWilliams et al., 2022). To minimise the costs associated with the phaseout of Russian energy imports, the European Commission launched the REPowerEU Plan, mobilising close to 300 billion euros through joint borrowing in the Recovery and Resilience Fund to diversify energy imports and encouraging investments in renewable energy (European Commission, 2022). This initiative likely had a positive impact on firm's investments, particularly for those with higher exposure to the loss of Russian energy imports. As illustrative firm-level evidence, Financial Times (2022) reports that TotalEnergies—which belongs to the high geoeconomic exposure group—announced higher net investment in oil, gas, and renewables as part of a growth plan excluding Russian activities.

> *"Pouyanné [CEO] said Total would* ***increase its net investments in oil, gas and renewable energy over the next three years****, as part of a* ***growth plan that excludes business interests in Russia****." – Financial Times, 28 September 2022.*

Similar considerations appear in the 2021/22 annual report of EVN AG's—another firm in the high geoeconomic exposure group—which describes steps to dispose of Russian-linked assets, secure non-Russian gas supplies, and expand renewable investment (EVN AG, 2022).

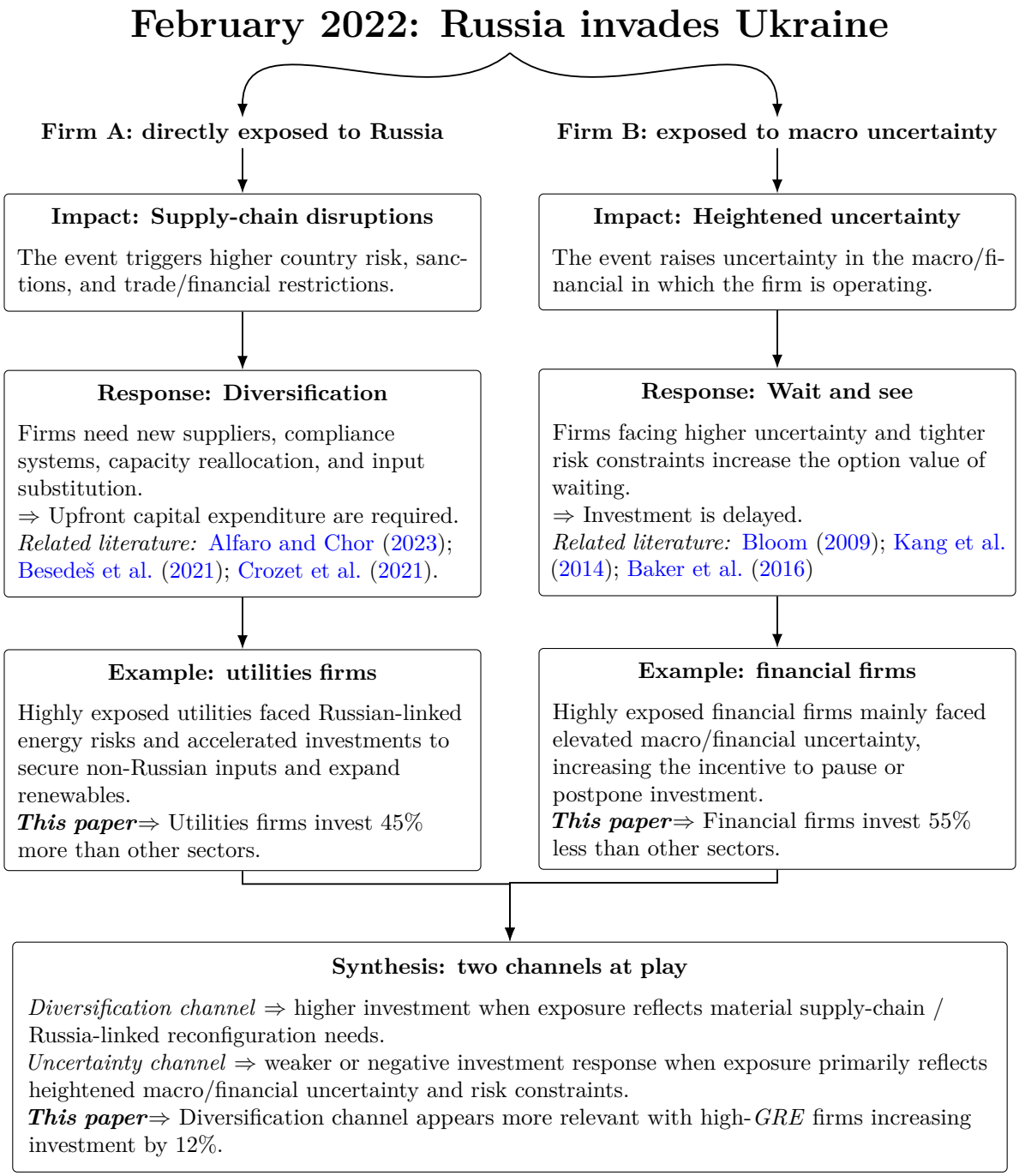
By contrast, highly exposed financial firms may have reduced investments more so than other sectors due to the sector's greater exposure to the heightened uncertainty that followed the war. While firms in other sectors may have been highly exposed due to their material exposure to Russia, financial firms may have been more likely to be exposed to the heightened uncertainty in the macroeconomic environment. In turn, highly exposed firms in the financial sector may have been more likely to pause investments, consistent with the literature on uncertainty and investments (Bloom, 2009). A schematic overview of the suggested distinct channels that may have affected firms' decision to invest, or not to invest, following Russia's invasion is shown in Figure 7.1.

7.2 Limitations: threats to identification, measurement, and external validity

A first limitation concerns threats to causal identification. While the difference-in-differences design is supported by extensive robustness checks—including firm fixed effects, event-study evidence of parallel trends, and placebo tests—one may object that firms are split between high and low exposure groups despite almost all firms being materially affected by the increase in geoeconomic risks following Russia's invasion of Ukraine. All firms being affected results in an econometric case where there are no "untreated units", and not accounting for this problem can lead to biased estimators (de Chaisemartin et al., 2025). However, the measurement of geoeconomic risk exposure through earnings calls makes the identification of non-treated units impossible given a large number of firms not mentioning geoeconomic risks in 2022.

Relatedly, a second limitation is concerned with the measurement of geoeconomic risk exposure. The *GRE* captures the salience of geoeconomic risks in corporate earnings calls,

Figure 7.1: Visual overview of proposed mechanisms



rather than firms' direct monetary exposure to sanctions, trade restrictions, or disrupted supply chains. The index also does not allow to separate between firms bringing up risks due to exposure to macroeconomic uncertainty effects and direct exposure to country-specific developments. This distinction would be especially relevant to distinguish the effects of the two channels proposed above. As such, the index reflects perceived and communicated risk, which may be influenced by a variety of unobserved reasons other than actual balance-sheet exposure. The measure also does not appear to be forward-looking in terms of identified dependencies, but rather reacting to events as they unfold.

A third limitation concerns external validity. The positive investment response of highly exposed firms is concentrated among EU-headquartered firms and does not generalise to firms outside the EU. This suggests that the estimated effects may be shaped by EU-specific policy environments, including energy policy responses such as the REPowerEU Plan discussed above, industrial support measures, and the EU's particular exposure to the Russo–Ukrainian war. Likewise, the analysis was conducted on publicly listed firms and it is unclear to what extent the findings could be extended to private firms. Consequently, the findings should not be interpreted as a universal firm-level response to geoeconomic risk, but rather as evidence of a context-dependent adjustment within the EU's economic and policy framework.

8 Conclusion

This paper has proposed a new index that tracks firms exposure to geoeconomic risks over time using corporate earnings call transcripts, the *GRE*, and evaluated the effects of high exposure on investments for EU firms following Russia's invasion of Ukraine.

I find that firm-level geoeconomic risk salience rises sharply after major geoeconomic events and varies systematically between different geographic headquarters and sectors. Using a Differences-in-Differences approach, I empirically confirm the descriptive evidence of increased investments for firms with above-median *GRE* in 2022, finding a baseline 12% increase in investments. This result holds up to a series of robustness checks and corresponds to about double the average annual investment growth rate for these firms. I show pronounced heterogeneity between the investment response of highly exposed firms in the utilities and

financial sectors relative to the rest. Firms in the utilities sector increase comparatively increase investments by 55%, while financial firms cut back on investments 45% more than firms in other sectors.

I propose a simple framework for firms' investment decision-making faced with geoeconomic risks with two channels that help rationalise the empirical results: a diversification, and an uncertainty channel. The diversification channel would explain the overall increase in investment among EU firms, reflecting the required upfront investments to friend-shore and diversify away from Russia. This channel is also consistent with the larger response among utilities firms, which were particularly exposed to Russian energy imports. In turn, the uncertainty channel would account for the investment decline among financial firms, which were more exposed to a fragile macroeconomic environment.

Interesting future avenues for research on this topic include providing causal evidence on the existence of the proposed channels explaining sectoral the differences in investment patterns, developing firm-level geoeconomic risk measures that combine text-based evidence with monetary exposures, and assessing the total adjustment cost to the Russian invasion of Ukraine for exposed firms.

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APPENDIX

A The Geoeconomic Risk Exposure index

This appendix provides more details on the earnings call data used for the construction of the Geoeconomic Risk Exposure (GRE) and on its results. First, the cleaning steps are discussed. Then, the dataset coverage for European firms is shown in Table A.1. Table A.2 provides the full list of keyword queries used to detect references geoeconomic threats in company transcripts. I split the terms by type of geoeconomic risk as shown in the main text. Lastly, Table A.3 shows the country ranking by GRE.

Cleaning steps. A few important cleaning steps needed to be followed in order to make the data usable for the analysis. Firstly I needed to restrict the data to only rows which had a gvkey value since this was the firm identifier used link the dataset with the Compustat data. This cleaning step resulted in 15,678 transcripts being deleted. Having removed unidentified firms, I needed to handle duplicated values at the firm-quarter level, again to ensure a one-to-one match with the firm-level data. Duplicates could be errors from the dataset, or firms holding more than one earning call per quarter. I collapse these duplicated observations to the mean value of the number of observed duplicates, which ensures that both errors and "genuine" duplicates are handled correctly and result in one observation per firm-quarter. Doing so results in a full dataset of 416,522 transcripts.

Table A.1: Dataset coverage by firm's headquarters country

Country	Firms	Share (%)	Transcripts	Share (%)
Austria	47	2.54	1,483	2.66
Belgium	74	4.00	1,887	3.39
Cyprus	22	1.19	303	0.54
Czech Republic	5	0.27	229	0.41
Denmark	77	4.16	3,246	5.83
Estonia	3	0.16	24	0.04
Finland	119	6.42	3,247	5.83
France	211	11.39	6,201	11.13
Germany	294	15.87	9,152	16.43
Greece	62	3.35	1,535	2.76
Hungary	4	0.22	244	0.44
Ireland	79	4.27	3,263	5.86
Italy	131	7.07	4,197	7.54
Lithuania	3	0.16	21	0.04
Luxembourg	41	2.21	1,432	2.57
Malta	5	0.27	90	0.16
Netherlands	123	6.64	3,974	7.13
Poland	48	2.59	1,216	2.18
Portugal	21	1.13	753	1.35
Romania	16	0.86	148	0.27
Slovenia	1	0.05	23	0.04
Spain	92	4.97	3,751	6.73
Sweden	374	20.19	9,281	16.66
Total (EU)	1,852	100.00	55,700	100.00

Notes: This Table shows the number of unique firms by headquarters country and their share in the full dataset, and the corresponding number and share of transcripts by headquarters country.

Table A.2: Search terms used in the text-mining query

Category	Query
Trade risks	(tariff* OR import dut* OR import barrier* OR trade treat* OR trade polic* OR trade act* OR (anti-)dumping OR trade agreement* OR trade relationship* OR GATT OR World Trade Organization/WTO OR free trade) AND (risk* OR threat* OR caution* OR uncertaint* OR propos* OR future OR worr* OR concern* OR volatile OR tension* OR likel* OR probab* OR possib* OR chance* OR danger* OR fear* OR expect* OR potential OR rumor* OR and prospect*) OR (import* OR export* OR border*) AND (ban* OR tax* OR subsid* OR control*) AND NOT (tariff AND (feed-in OR MTA OR network* OR transportation OR adjustment* OR regulat* OR rate* OR escalator))
Sanctions risks	(sanctions OR asset freez*) AND (risk* OR threat* OR caution* OR uncertaint* OR propos* OR future OR worr* OR concern* OR volatile OR tension* OR likel* OR probab* OR possib* OR chance* OR danger* OR fear* OR expect* OR potential OR rumor* OR and prospect*)
Embargo risks	embargo* AND (risk* OR threat* OR caution* OR uncertaint* OR propos* OR future OR worr* OR concern* OR volatile OR tension* OR likel* OR probab* OR possib* OR chance* OR danger* OR fear* OR expect* OR potential OR rumor* OR and prospect*)
Geopolitical risks	(((war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR revolt OR coup OR geopolitical) AND (threat* OR warn* OR fear* OR risk* OR concern* OR danger* OR doubt* OR crisis OR trouble* OR dispute* OR tension* OR imminent* OR inevitable OR footing OR menace* OR brink OR scare OR peril*)) OR ((peace OR truce OR armistice OR treaty OR parley) AND (threat* OR menace* OR reject* OR peril* OR boycott* OR disrupt*)) OR ((military OR troops OR missile* OR "arms" OR weapon* OR bomb* OR warhead*) AND (buildup* OR build-up* OR sanction* OR blockade* OR embargo OR quarantine OR ultimatum OR mobilize*)) OR ((terror* OR guerrilla* OR hostage*) AND (threat* OR warn* OR fear* OR risk* OR concern* OR danger* OR doubt* OR crisis OR trouble* OR dispute* OR tension* OR imminent* OR inevitable OR footing OR menace* OR brink OR scare OR peril*)) OR ((allies* OR enemy* OR insurgent* OR foe* OR army OR navy OR aerial OR troops OR rebels) AND (advance* OR attack* OR strike* OR drive* OR shell* OR offensive OR invasion OR invade* OR clash* OR raid* OR launch*)) OR ((war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR revolt OR coup OR geopolitical) AND (begin* OR start* OR declar* OR begun OR began OR outbreak OR "broke out" OR breakout OR proclamation OR launch*)))

Notes: This table shows the query run at the sentence level on all available transcripts to identify mentions of geoeconomic risk exposure. The query search terms are taken from Caldara and Iacoviello (2022) and Caldara et al. (2020), and Niepmann and Shen (2024) complemented with geoeconomic terms related to the LLM query in Clayton et al. (2025a). An "*" allows zero or more arbitrary characters until the next word boundary (e.g. missile* counts both *missile* and *missiles*).

Table A.3: GRE by country

Rank	Country	Firms	Total exposure	Exposure per 1,000
1	Slovenia	1	53	3.139
2	Portugal	21	827	2.879
3	Lithuania	3	16	2.727
4	Estonia	3	14	2.533
5	Romania	16	98	2.221
6	Poland	48	1,031	2.205
7	Hungary	4	226	2.127
8	Czech Republic	5	194	2.113
9	Italy	131	2,910	1.687
10	Cyprus	22	149	1.516
11	Greece	62	712	1.501
12	Spain	92	2,032	1.301
13	Austria	47	682	1.160
14	Luxembourg	41	665	1.106
15	Finland	119	999	0.884
16	Belgium	74	757	0.878
17	Germany	294	3,708	0.860
18	Netherlands	123	1,548	0.786
19	France	211	2,449	0.764
20	Sweden	374	2,470	0.719
21	Denmark	77	910	0.691
22	Ireland	79	919	0.613
23	Malta	5	17	0.466
	Total (EU)	1,852	23,386	1.516

Notes: The Table shows the total and average geoeconomic risk exposure for each country over the entire time period, and the number of firms for each country.

B Methodology

This Appendix shows the cleaning steps of the Compustat data, the results of the t-test between firm's with high and low geoeconomic risk exposure (Table B.1), and the summary statistics (Table B.2).

Cleaning steps. To prepare the Compustat data for the analysis, I remove any duplicates based on repeated observations at the quarterly level per firm (removes 0.3% of observations). Next, using the exchange rate provided by Compustat, I convert all variables of interest into euros. I deflate all variables of interest after having merged in CPI and other country-level data from Global Macro Database. I create the log values for those variables of interest, and compute additional ratios of interest such as the cashflow, or the profit margin.

Table B.1: T-test results

	High exposure			Low exposure			T-Test
Variable	Mean	P25	P75	Mean	P25	P75	Difference
Number of firms	316			330			-14
Equity / assets	0.394	0.272	0.513	0.377	0.263	0.509	0.017***
Cashflow	-0.004	-0.019	0.004	-0.001	-0.020	0.003	-0.003*
Total assets	9.271	7.605	10.831	9.335	7.601	11.004	-0.064**
Sales	6.988	5.623	8.324	7.297	5.624	8.985	-0.309***
COGS	6.500	4.960	7.868	6.704	4.879	8.569	-0.203***
Debt	5.837	4.091	7.570	5.835	3.952	7.670	0.002
PPE	7.057	5.360	8.685	7.175	5.425	8.755	-0.118***
Interest exp.	3.106	1.464	4.696	3.156	1.457	4.773	-0.050*

Notes: This Table shows the mean, 25th and 75th percentiles of listed balance sheet variables for firms in the treatment (high exposure) and control (low exposure) groups. The last column reports the t-test difference between the two groups. All balance sheet variables which are not ratios are in logs, and their initial unit is in millions. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

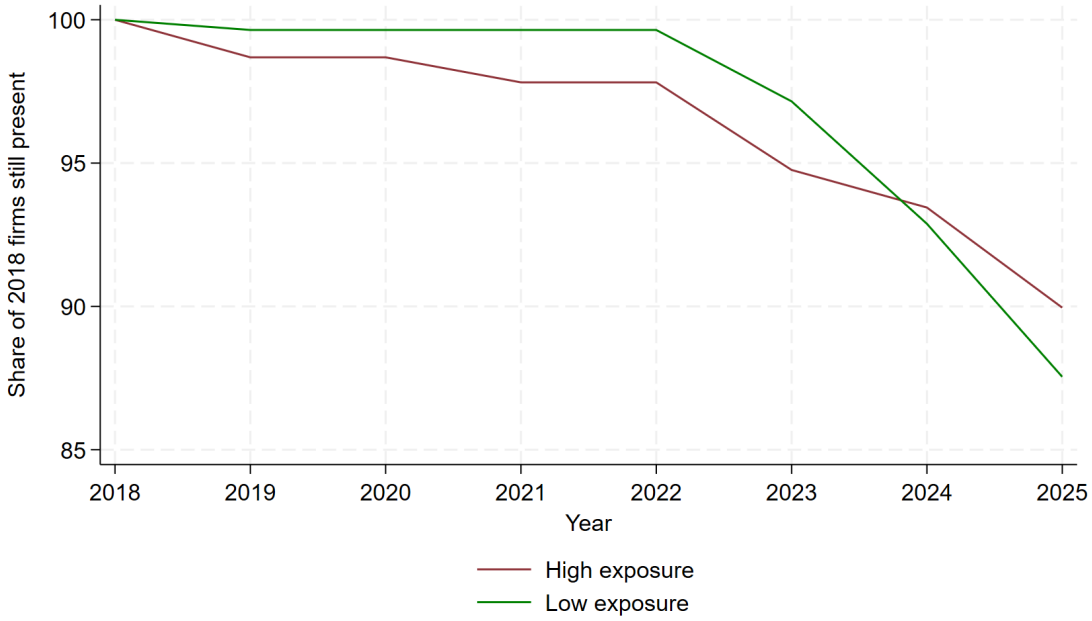
Table B.2: Summary statistics

Variable	N	Mean	p25	p75	Min	Max	Source
Investment ratio (CAPEX / assets)	27948	0.0104	0.0037	0.0141	-3.8668	6.3938	Compustat
Firm age (years)	36504	17.8904	17.0000	20.0000	3.0000	22.0000	Compustat
Cash flow (ratio)	28690	-0.0026	-0.0199	0.0037	-10.8279	9.4175	Compustat
Equity-to-assets ratio	29624	0.3847	0.2681	0.5107	-22.5096	1.1732	Compustat
Equity-to-leverage (ratio)	29064	34.0008	0.7684	2.8034	-78.1955	204291.8281	Compustat
Financing (net, scaled)	28686	0.0099	-0.0344	0.0062	-39.0000	50.5298	Compustat
Leverage (ratio)	35185	0.3143	0.1456	0.3861	0.0000	269.3620	Compustat
Log real total assets (EUR)	34354	9.3030	7.6033	10.9208	-0.1347	22.9725	Compustat
Log real cost of goods sold (EUR)	27861	6.6077	4.9206	8.2424	-8.1039	20.1182	Compustat
Log real total debt (EUR)	33525	5.8357	4.0318	7.6217	-6.9078	19.7025	Compustat
Log real dividends (EUR)	9909	2.8494	1.0513	5.0951	-10.1557	17.7461	Compustat
Log real intangible assets (EUR)	33473	6.7243	5.0430	8.4798	-7.5888	18.7864	Compustat
Log real PP&E (EUR)	32625	7.1153	5.3923	8.7257	-6.6454	20.7348	Compustat
Log real sales (EUR)	28379	7.1511	5.6236	8.6411	-7.8240	20.4373	Compustat
Log real interest expense (EUR)	34008	3.1313	1.4586	4.7392	-10.2537	18.3799	Compustat
Profit margin (ratio)	29583	-0.1739	0.2278	0.5567	-2525.3501	1469.7111	Compustat
Central bank policy rate (%)	36456	0.7475	-0.4000	1.2500	-0.7500	13.0000	Global Macro Database
Inflation rate (%)	36504	2.2194	0.8200	2.5572	-2.0985	19.7050	Global Macro Database

Notes: This Table shows the number of observations (N), mean, 25th percentile, 75th percentile, min, max, and source for the firm-level balance sheet data from Compustat, and the additional controls for country-level data taken from the Global Macro Database. All firm-level variables which are not ratios have been deflated (as described in the cleaning process above), and are in logs, with their initial unit being in millions.

C Robustness

Figure C.1: Risk of attrition: share of surviving firms



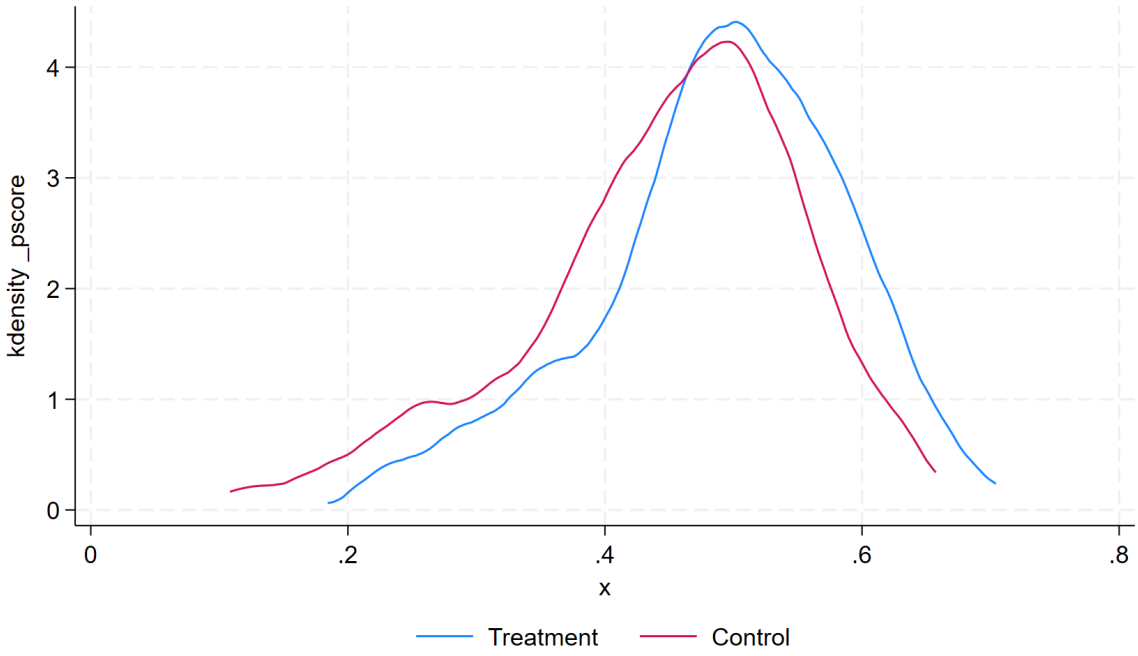
Notes: This Figure plots the share of firms in the dataset which have a recorded earnings call in 2018 which continue to be present in the dataset in the subsequent years. No new firms entering the sample after 2018 are considered in this Figure.

Table C.1: Controls interacted with post-invasion dummy

	(1)	(2)
D_i × Post-invasion	0.121**	0.113**
	(0.048)	(0.047)
Equity-to-assets	0.229	0.215
	(0.146)	(0.143)
Cash flow	-0.040	0.896***
	(0.124)	(0.191)
Log assets	0.308***	0.294***
	(0.087)	(0.089)
Log sales	0.291***	0.298***
	(0.059)	(0.065)
Log COGS	0.011	0.010
	(0.034)	(0.039)
Log debt	-0.035***	-0.034***
	(0.009)	(0.009)
Log PPE	0.368***	0.376***
	(0.067)	(0.069)
Log interest exp.	-0.049***	-0.044***
	(0.012)	(0.013)
Inflation	0.008	0.013
	(0.011)	(0.018)
CB rate	0.015	-0.004
	(0.019)	(0.033)
Equity-to-assets × Post-invasion		0.002
		(0.179)
Cash flow × Post-invasion		-1.065***
		(0.194)
Log assets × Post-invasion		0.003
		(0.059)
Log sales × Post-invasion		-0.001
		(0.080)
Log COGS × Post-invasion		-0.003
		(0.047)
Log debt × Post-invasion		-0.042**
		(0.020)
Log PPE × Post-invasion		0.081**
		(0.034)
Log interest exp × Post-invasion		-0.010
		(0.024)
Inflation × Post-invasion		-0.019
		(0.018)
CB rate × Post-invasion		0.013
		(0.029)
Observations	24,531	24,531
Firm FE	Yes	Yes
Quarter FE	Yes	Yes
Year FE	Yes	Yes

Notes: This Table shows the results of the baseline regression from estimating equation 4.1 in column (1) next to the baseline regression with all controls interacted with the post-invasion dummy (2). All firm-level controls are lagged. Robust standard errors clustered at the firm level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure C.2: Kernel density after PSM



Notes: This Figure depicts kernel density pscore for treatment and control groups after the Propensity Score Matching (PSM) weighting. For the PSM I use the pre-2022 averages for equity-to-assets, cash flow, log total assets, log sales, log cost of goods sold, log debt, log plant, property, and equipment, log total interest expenditures, and trim the dataset to remove the 5% of firms for which the pscore obatined by running the matching procedure is the lowest.